Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: PH Ranch

Physical address of dairy:

Number and Street	City	County	Zip Code
6335 W Oakdale RD	Winton	Merced	95388

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1955

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X140-X020-X010-XXXX X140-X020-X013-XXXX

B. OPERATORS

Van Waremerdam, Paul

Operator name: Van Waremerdam, Paul

Telephone no.: (123) 000-0000 Landline (123) 000-0000 Cellular

Mailing Address Number and Street	City	State	Zip Code
6335 Oakdale RD	Winton	CA	95388

This operator is responsible for paying permit fees.

C. OWNERS

Veldhuis, Jeanette

Legal owner name: Veldhuis, Jeanette

Telephone no.: (123) 000-0000 Landline Cellular

Mailing Address Number and Street	City	State	Zip Code
6335 Oakdale RD	Winton	CA	95388

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	400	400
Number under roof	2,300	230	1,400	1,200	0	0
Maximum number	2,300	230	1,500	1,300	450	450
Average number	2,300	230	1,400	1,200	450	450
Avg live weight (lbs)	1,400	1,500	100	850		

Predominant milk cow breed: Holstein

Average milk production: 95 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 94,764.10 *tons per reporting period*

Total nitrogen from manure: 1,194,310.68 *lbs per reporting period*

After ammonia losses (30% loss applied): 836,017.48 *lbs per reporting period*

Total phosphorus from manure: 197,361.21 *lbs per reporting period*

Total potassium from manure: 478,248.72 *lbs per reporting period*

Total salt from manure: 1,135,843.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 135,144,000 *gallons*

Total nitrogen generated: 249,751.76 *lbs*

Total phosphorus generated: 58,949.37 *lbs*

Total potassium generated: 301,328.47 *lbs*

Total salt generated: 1,545,552.75 *lbs*

	135,144,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	135,144,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #8	Ground water
Ag Well 4A East South	Ground water
Ag Well 4A West	Ground water
Ag Well 4B	Ground water
Canal Water	Surface water

Source Description	Type
Domestic Well #1	Ground water
Domestic Well #2	Ground water
Domestic Well #3 by Corral	Ground water
Domestic well #4 by Calf	Ground water
River Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/05/2023	Compost	3,563.22 *ton*	Dry-weight	44.9		23,200.00	7,770.00	20,200.00		38.70
10/07/2023	Corral solids	111.06 *ton*	Dry-weight	47.0		0.00	0.00	0.00		0.00
11/09/2023	Corral solids	122.54 *ton*	Dry-weight	36.1		0.00	0.00	0.00		0.00
11/14/2023	Corral solids	179.87 *ton*	Dry-weight	22.2		0.00	0.00	0.00		0.00
11/30/2023	Compost	2,327.81 *ton*	Dry-weight	44.9		23,200.00	7,770.00	20,200.00		38.70

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	150,612.43	50,442.18	131,136.68	2,512,371.13
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	150,612.43	50,442.18	131,136.68	2,512,371.13

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	16	16	2	process wastewater	X140-X020-X013-XXXX
2	50	50	2	process wastewater	X042-X140-X005-XXXX X140-X200-X013-XXXX
3	43	43	2	process wastewater	X140-X020-X013-XXXX
4	95	95	2	both	X140-X020-X004-XXXX X140-X020-X010-XXXX X140-X150-X032-XXXX X140-X150-X033-XXXX
4B	40	40	2	both	X140-X020-X010-XXXX X140-X150-X002-XXXX
5A	38	38	1	process wastewater	X042-X140-X005-XXXX
5B	16	16	2	process wastewater	X042-X140-X005-XXXX
5C	16	16	2	process wastewater	X042-X140-X005-XXXX X140-X020-X013-XXXX
6	105	105	2	process wastewater	X042-X140-X005-XXXX X140-X020-X003-XXXX X140-X020-X013-XXXX
7	128	128	2	process wastewater	X140-X020-X003-XXXX X140-X020-X004-XXXX X140-X020-X010-XXXX X140-X020-X013-XXXX
8	68	68	2	process wastewater	X042-X140-X005-XXXX X140-X020-X003-XXXX
9	45	45	2	process wastewater	X042-X140-X005-XXXX X140-X020-X003-XXXX
PH 24	20	20	0	none	X140-X002-X013-XXXX
PH 26	40	40	0	none	X140-X002-X013-XXXX
PH 27	20	20	0	none	X140-X002-X010-XXXX
PH 28	55	55	0	none	X140-X015-X033-XXXX
PS 29A	20	20	0	none	X140-X130-X015-XXXX X140-X130-X016-XXXX
PS 29B	20	20	0	none	X140-X130-X013-XXXX X140-X130-X014-XXXX
PS 30	40	40	0	none	X052-X090-X012-XXXX

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
WC-01	44	44	0	none	X052-X003-X039-XXXX X052-X003-X057-XXXX
WC-02	34	34	0	none	X052-X003-X040-XXXX
Totals for areas that were used for application	660	660	23		
Totals for areas that were not used for application	293	293	0		
Land application area totals	953	953	23		

B. CROPS AND HARVESTS

1

Field name: 1

11/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 16 Plant date: 11/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	192.74 *ton*	Dry-weight		55.5	13,900.00	1,760.00	17,100.00		14.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.05	149.02	18.87	183.33	1,543.85

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	437.46 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.34	0.00	0.00	0.00	0.00

2

Field name: 2

2

11/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 50 Plant date: 11/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	644.60 *ton*	Dry-weight		55.6	20,300.00	3,210.00	31,400.00		17.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.89	232.40	36.75	359.47	2,037.76

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,652.65 *ton*	Dry-weight		71.6	13,100.00	1,990.00	13,000.00		2.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.05	245.94	37.36	244.06	407.40

3

Field name: 3

11/29/2022: Winter Forage

Crop: Winter Forage Acres planted: 43 Plant date: 11/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	504.73 *ton*	Dry-weight		65.3	14,400.00	1,800.00	18,800.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.74	117.30	14.66	153.15	1,132.31

3

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	1,568.39 *ton*	Dry-weight		70.7	12,100.00	2,150.00	11,300.00		2.04

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	36.47	258.62	45.95	241.52	436.03

4

Field name: 4

11/22/2022: Winter Forage

Crop: Winter Forage Acres planted: 95 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	1,238.32 *ton*	Dry-weight		62.8	14,000.00	1,730.00	17,500.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.03	135.77	16.78	169.72	1,260.74

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 95 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	2,789.59 *ton*	Dry-weight		65.3	7,000.00	1,740.00	7,930.00		2.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.36	142.65	35.46	161.60	442.22

4B

Field name: 4B

11/21/2022: Winter Forage

Crop: Winter Forage Acres planted: 40 Plant date: 11/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	455.58 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.39	0.00	0.00	0.00	0.00

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	959.88 *ton*	Dry-weight		69.2	10,600.00	2,530.00	8,190.00		2.81

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.00	156.69	37.40	121.07	415.38

5A

Field name: 5A

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	1,103.00 *ton*	Dry-weight		67.7	18,700.00	2,600.00	15,700.00		1.96

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.03	350.64	48.75	294.39	367.52

5B

Field name: 5B

11/23/2022: Winter Forage

Crop: Winter Forage Acres planted: 16 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	233.77 *ton*	Dry-weight		52.3	13,700.00	1,870.00	17,400.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.61	190.96	26.07	242.53	1,714.44

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	486.22 *ton*	Dry-weight		62.7	14,600.00	2,510.00	9,690.00		8.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.39	330.98	56.90	219.67	1,915.62

5C

Field name: 5C

11/17/2022: Winter Forage

Crop: Winter Forage Acres planted: 16 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	287.19 *ton*	Dry-weight		49.1	15,100.00	2,170.00	19,300.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.95	275.91	39.65	352.66	2,521.60

5C

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	545.57 *ton*	Dry-weight		66.5	21,100.00	2,130.00	9,680.00		2.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.10	482.05	48.66	221.15	470.62

6

Field name: 6

11/30/2022: Winter Forage

Crop: Winter Forage Acres planted: 105 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/14/2023	1,210.81 *ton*	Dry-weight		49.6	22,300.00	1,960.00	17,800.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.53	259.21	22.78	206.90	1,336.73

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 105 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	3,695.35 *ton*	Dry-weight		64.6	12,700.00	2,380.00	9,660.00		1.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.19	316.45	59.30	240.70	408.64

7

Field name: 7

11/18/2022: Winter Forage

Crop: Winter Forage Acres planted: 128 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	2,209.50 *ton*	Dry-weight		52.5	10,700.00	1,280.00	14,300.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.26	175.47	20.99	234.50	1,475.88

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 128 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	3,919.27 *ton*	Dry-weight		68.2	25,800.00	2,240.00	14,000.00		3.24

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.62	502.43	43.62	272.63	630.95

8

Field name: 8

11/22/2022: Winter Forage

Crop: Winter Forage Acres planted: 68 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	855.71 *ton*	Dry-weight		61.1	9,490.00	1,580.00	11,800.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.58	92.91	15.47	115.53	851.76

8

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 68 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	2,436.00 *ton*	Dry-weight		71.8	14,300.00	1,520.00	12,900.00		2.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.82	288.92	30.71	260.64	484.91

9

Field name: 9

11/22/2022: Winter Forage

Crop: Winter Forage Acres planted: 45 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	615.69 *ton*	Dry-weight		75.9	24,600.00	2,960.00	26,300.00		22.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.68	162.23	19.52	173.44	1,477.22

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	1,401.22 *ton*	Dry-weight		70.6	19,600.00	2,300.00	10,800.00		1.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.14	358.86	42.11	197.74	325.91

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/28/2022: Winter Forage

Field name: 1

Crop: Winter Forage Plant date: 11/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.51	21.03	130.49	675.95	360,000.00 *gal*
Application event totals		99.51	21.03	130.49	675.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.51	21.03	130.49	675.95	360,000.00 *gal*
Application event totals		99.51	21.03	130.49	675.95	

1 - 06/15/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.93	24.51	111.76	576.81	1,152,000.00 *gal*
River Water	Surface water	0.44	0.00	0.00	45.96	1,836,000.00 *gal*
Application event totals		92.37	24.51	111.76	622.77	

1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.67	26.05	118.74	612.86	1,224,000.00 *gal*
River Water	Surface water	0.44	0.00	0.00	45.96	1,836,000.00 *gal*
Application event totals		98.11	26.05	118.74	658.82	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.95	18.39	83.82	432.60	864,000.00 *gal*
River Water	Surface water	0.44	0.00	0.00	45.96	1,836,000.00 *gal*
Application event totals		69.39	18.39	83.82	478.57	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.95	18.39	83.82	432.60	864,000.00 *gal*
River Water	Surface water	0.44	0.00	0.00	45.96	1,836,000.00 *gal*
Application event totals		69.39	18.39	83.82	478.57	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.96	12.26	55.88	288.40	576,000.00 *gal*
River Water	Surface water	0.44	0.00	0.00	45.96	1,836,000.00 *gal*
Application event totals		46.40	12.26	55.88	334.37	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.44	0.00	0.00	45.96	1,836,000.00 *gal*
Application event totals		0.44	0.00	0.00	45.96	

1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.44	0.00	0.00	45.96	1,836,000.00 *gal*
Application event totals		0.44	0.00	0.00	45.96	

2 - 11/28/2022: Winter Forage

Field name: 2

Crop: Winter Forage Plant date: 11/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.69	13.46	83.52	432.60	720,000.00 *gal*
Application event totals		63.69	13.46	83.52	432.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.69	13.46	83.52	432.60	720,000.00 *gal*
Application event totals		63.69	13.46	83.52	432.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.69	13.46	83.52	432.60	720,000.00 *gal*
River Water	Surface water	0.04	0.00	0.00	4.33	540,000.00 *gal*
Application event totals		63.73	13.46	83.52	436.93	

2 - 06/15/2023: Corn, silage

Field name: 2

Crop: Corn, silage Plant date: 06/15/2023

2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.77	9.81	44.70	230.72	1,440,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		37.17	9.81	44.70	272.25	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.42	7.84	35.76	184.58	1,152,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		29.82	7.84	35.76	226.11	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.42	7.84	35.76	184.58	1,152,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		29.82	7.84	35.76	226.11	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.42	7.84	35.76	184.58	1,152,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		29.82	7.84	35.76	226.11	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.06	5.88	26.82	138.43	864,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		22.46	5.88	26.82	179.96	

2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.06	5.88	26.82	138.43	864,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		22.46	5.88	26.82	179.96	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.06	5.88	26.82	138.43	864,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		22.46	5.88	26.82	179.96	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	127.38	9.48	41.58	240.34	720,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		127.78	9.48	41.58	281.87	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		0.40	0.00	0.00	41.53	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.40	0.00	0.00	41.53	5,184,000.00 *gal*
Application event totals		0.40	0.00	0.00	41.53	

3 - 11/29/2022: Winter Forage

Field name: 3

Crop: Winter Forage

Plant date: 11/29/2022

3 - 11/29/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.06	15.65	97.11	503.03	720,000.00 *gal*
Application event totals		74.06	15.65	97.11	503.03	

01/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.06	15.65	97.11	503.03	720,000.00 *gal*
Application event totals		74.06	15.65	97.11	503.03	

02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.05	0.00	0.00	5.03	540,000.00 *gal*
Application event totals		0.05	0.00	0.00	5.03	

3 - 06/16/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.48	10.26	46.78	241.45	1,296,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		38.89	10.26	46.78	283.71	

3 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.93	7.98	36.39	187.80	1,008,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		30.34	7.98	36.39	230.05	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.93	7.98	36.39	187.80	1,008,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		30.34	7.98	36.39	230.05	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.93	7.98	36.39	187.80	1,008,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		30.34	7.98	36.39	230.05	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	21.38	5.70	25.99	134.14	720,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		21.78	5.70	25.99	176.40	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	21.38	5.70	25.99	134.14	720,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		21.78	5.70	25.99	176.40	

3 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	148.11	11.02	48.35	279.46	720,000.00 *gal*
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		148.52	11.02	48.35	321.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		0.40	0.00	0.00	42.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.40	0.00	0.00	42.25	4,536,000.00 *gal*
Application event totals		0.40	0.00	0.00	42.25	

4 - 11/22/2022: Winter Forage

Field name: 4

Crop: Winter Forage

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.41	2.83	17.58	91.07	288,000.00 *gal*
Application event totals		13.41	2.83	17.58	91.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost Manure	Compost	127.83	42.81	111.30	2,132.37	475.00 *ton*
Application event totals		127.83	42.81	111.30	2,132.37	

4 - 11/22/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.41	2.83	17.58	91.07	288,000.00 *gal*
Application event totals		13.41	2.83	17.58	91.07	

03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	14.70	2.99	19.58	70.84	288,000.00 *gal*
Application event totals		14.70	2.99	19.58	70.84	

4 - 06/19/2023: Corn, silage

Field name: 4

Crop: Corn, silage Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	23.22	6.19	28.23	145.72	1,728,000.00 *gal*
Ag Well 4A East South	Ground water	11.70	0.00	0.00	359.45	3,720,000.00 *gal*
Ag Well 4A West	Ground water	9.21	0.00	0.00	980.32	3,720,000.00 *gal*
Application event totals		44.14	6.19	28.23	1,485.49	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4A East South	Ground water	11.70	0.00	0.00	359.45	3,720,000.00 *gal*
Ag Well 4A West	Ground water	9.21	0.00	0.00	980.32	3,720,000.00 *gal*
Application event totals		20.91	0.00	0.00	1,339.77	

4 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	23.22	6.19	28.23	145.72	1,728,000.00 *gal*
Ag Well 4A East South	Ground water	9.43	0.00	0.00	289.88	3,000,000.00 *gal*
Ag Well 4A West	Ground water	7.43	0.00	0.00	790.58	3,000,000.00 *gal*
Application event totals		40.09	6.19	28.23	1,226.18	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4A East South	Ground water	9.43	0.00	0.00	289.88	3,000,000.00 *gal*
Ag Well 4A West	Ground water	7.43	0.00	0.00	790.58	3,000,000.00 *gal*
Application event totals		16.87	0.00	0.00	1,080.46	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4A East South	Ground water	9.43	0.00	0.00	289.88	3,000,000.00 *gal*
Ag Well 4A West	Ground water	7.43	0.00	0.00	790.58	3,000,000.00 *gal*
Application event totals		16.87	0.00	0.00	1,080.46	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4A East South	Ground water	9.43	0.00	0.00	289.88	3,000,000.00 *gal*
Ag Well 4A West	Ground water	7.43	0.00	0.00	790.58	3,000,000.00 *gal*
Application event totals		16.87	0.00	0.00	1,080.46	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4A East South	Ground water	9.43	0.00	0.00	289.88	3,000,000.00 *gal*
Ag Well 4A West	Ground water	7.43	0.00	0.00	790.58	3,000,000.00 *gal*
Application event totals		16.87	0.00	0.00	1,080.46	

4 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4A East South	Ground water	9.43	0.00	0.00	289.88	3,000,000.00 *gal*
Ag Well 4A West	Ground water	7.43	0.00	0.00	790.58	3,000,000.00 *gal*
Application event totals		16.87	0.00	0.00	1,080.46	

4B - 11/21/2022: Winter Forage

Field name: 4B

Crop: Winter Forage Plant date: 11/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.84	6.73	41.76	216.30	288,000.00 *gal*
Application event totals		31.84	6.73	41.76	216.30	

11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost Manure	Compost	127.83	42.81	111.30	2,132.37	200.00 *ton*
Application event totals		127.83	42.81	111.30	2,132.37	

01/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.84	6.73	41.76	216.30	288,000.00 *gal*
Application event totals		31.84	6.73	41.76	216.30	

4B - 06/19/2023: Corn, silage

Field name: 4B

Crop: Corn, silage Plant date: 06/19/2023

4B - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.58	7.35	33.53	173.04	864,000.00 *gal*
Ag Well 4B	Ground water	14.86	0.00	0.00	454.24	4,536,000.00 *gal*
Application event totals		42.44	7.35	33.53	627.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4B	Ground water	14.86	0.00	0.00	454.24	4,536,000.00 *gal*
Application event totals		14.86	0.00	0.00	454.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.58	7.35	33.53	173.04	864,000.00 *gal*
Ag Well 4B	Ground water	14.86	0.00	0.00	454.24	4,536,000.00 *gal*
Application event totals		42.44	7.35	33.53	627.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4B	Ground water	12.73	0.00	0.00	389.34	3,888,000.00 *gal*
Application event totals		12.73	0.00	0.00	389.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	18.39	4.90	22.35	115.36	576,000.00 *gal*
Ag Well 4B	Ground water	12.73	0.00	0.00	389.34	3,888,000.00 *gal*
Application event totals		31.12	4.90	22.35	504.71	

4B - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4B	Ground water	12.73	0.00	0.00	389.34	3,888,000.00 *gal*
Application event totals		12.73	0.00	0.00	389.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4B	Ground water	12.73	0.00	0.00	389.34	3,888,000.00 *gal*
Application event totals		12.73	0.00	0.00	389.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4B	Ground water	12.73	0.00	0.00	389.34	3,888,000.00 *gal*
Application event totals		12.73	0.00	0.00	389.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4B	Ground water	12.73	0.00	0.00	389.34	3,888,000.00 *gal*
Application event totals		12.73	0.00	0.00	389.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 4B	Ground water	12.73	0.00	0.00	389.34	3,888,000.00 *gal*
Application event totals		12.73	0.00	0.00	389.34	

5A - 06/13/2023: Corn, silage

Field name: 5A

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5A - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.55	11.61	52.94	273.22	1,296,000.00 *gal*
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		43.94	11.61	52.94	314.21	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.71	10.32	47.06	242.87	1,152,000.00 *gal*
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		39.10	10.32	47.06	283.85	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.71	10.32	47.06	242.87	1,152,000.00 *gal*
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		39.10	10.32	47.06	283.85	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.71	10.32	47.06	242.87	1,152,000.00 *gal*
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		39.10	10.32	47.06	283.85	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.03	7.74	35.29	182.15	864,000.00 *gal*
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		29.42	7.74	35.29	223.13	

5A - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.03	7.74	35.29	182.15	864,000.00 *gal*
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		29.42	7.74	35.29	223.13	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.03	7.74	35.29	182.15	864,000.00 *gal*
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		29.42	7.74	35.29	223.13	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.03	7.74	35.29	182.15	864,000.00 *gal*
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		29.42	7.74	35.29	223.13	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		0.39	0.00	0.00	40.98	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.39	0.00	0.00	40.98	3,888,000.00 *gal*
Application event totals		0.39	0.00	0.00	40.98	

5B - 11/23/2022: Winter Forage

Field name: 5B

Crop: Winter Forage

Plant date: 11/23/2022

5B - 11/23/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	139.32	29.44	182.69	946.32	504,000.00 *gal*
Application event totals		139.32	29.44	182.69	946.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	119.42	25.24	156.59	811.13	432,000.00 *gal*
Application event totals		119.42	25.24	156.59	811.13	

5B - 06/13/2023: Corn, silage

Field name: 5B

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.96	12.26	55.88	288.40	576,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		46.22	12.26	55.88	315.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
River Water	Surface water	0.10	0.00	0.00	10.82	432,000.00 *gal*
Application event totals		0.36	0.00	0.00	37.85	

5B - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.47	9.19	41.91	216.30	432,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		34.73	9.19	41.91	243.34	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.47	9.19	41.91	216.30	432,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		34.73	9.19	41.91	243.34	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.47	9.19	41.91	216.30	432,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		34.73	9.19	41.91	243.34	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.98	6.13	27.94	144.20	288,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		23.24	6.13	27.94	171.24	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.98	6.13	27.94	144.20	288,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		23.24	6.13	27.94	171.24	

5B - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.98	6.13	27.94	144.20	288,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		23.24	6.13	27.94	171.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		0.26	0.00	0.00	27.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		0.26	0.00	0.00	27.04	

5C - 11/17/2022: Winter Forage

Field name: 5C

Crop: Winter Forage

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.51	21.03	130.49	675.95	360,000.00 *gal*
Application event totals		99.51	21.03	130.49	675.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.61	16.82	104.40	540.76	288,000.00 *gal*
Application event totals		79.61	16.82	104.40	540.76	

5C - 11/17/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.61	16.82	104.40	540.76	288,000.00 *gal*
Application event totals		79.61	16.82	104.40	540.76	

5C - 06/14/2023: Corn, silage

Field name: 5C

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.96	12.26	55.88	288.40	576,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		46.22	12.26	55.88	315.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.47	9.19	41.91	216.30	432,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		34.73	9.19	41.91	243.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.47	9.19	41.91	216.30	432,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		34.73	9.19	41.91	243.34	

5C - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.47	9.19	41.91	216.30	432,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		34.73	9.19	41.91	243.34	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.47	9.19	41.91	216.30	432,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		34.73	9.19	41.91	243.34	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.98	6.13	27.94	144.20	288,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		23.24	6.13	27.94	171.24	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.98	6.13	27.94	144.20	288,000.00 *gal*
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		23.24	6.13	27.94	171.24	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		0.26	0.00	0.00	27.04	

5C - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.26	0.00	0.00	27.04	1,080,000.00 *gal*
Application event totals		0.26	0.00	0.00	27.04	

6 - 11/30/2022: Winter Forage

Field name: 6

Crop: Winter Forage

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.52	10.25	63.63	329.60	1,152,000.00 *gal*
Application event totals		48.52	10.25	63.63	329.60	

01/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.52	10.25	63.63	329.60	1,152,000.00 *gal*
Application event totals		48.52	10.25	63.63	329.60	

02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.52	10.25	63.63	329.60	1,152,000.00 *gal*
Application event totals		48.52	10.25	63.63	329.60	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.19	10.80	70.86	256.36	1,152,000.00 *gal*
Application event totals		53.19	10.80	70.86	256.36	

6 - 06/13/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.04	16.81	76.63	395.52	5,184,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.27	16,848,000.00 *gal*
Application event totals		63.65	16.81	76.63	459.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.03	14.94	68.12	351.58	4,608,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.27	16,848,000.00 *gal*
Application event totals		56.65	14.94	68.12	415.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.53	14.01	63.86	329.60	4,320,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.27	16,848,000.00 *gal*
Application event totals		53.15	14.01	63.86	393.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.53	14.01	63.86	329.60	4,320,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.27	16,848,000.00 *gal*
Application event totals		53.15	14.01	63.86	393.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.53	14.01	63.86	329.60	4,320,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.27	16,848,000.00 *gal*
Application event totals		53.15	14.01	63.86	393.88	

6 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.62	0.00	0.00	64.27	16,848,000.00 *gal*
Application event totals		0.62	0.00	0.00	64.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.62	0.00	0.00	64.27	16,848,000.00 *gal*
Application event totals		0.62	0.00	0.00	64.27	

7 - 11/18/2022: Winter Forage

Field name: 7

Crop: Winter Forage

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.71	12.62	78.30	405.57	1,728,000.00 *gal*
Application event totals		59.71	12.62	78.30	405.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.71	12.62	78.30	405.57	1,728,000.00 *gal*
Application event totals		59.71	12.62	78.30	405.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.71	12.62	78.30	405.57	1,728,000.00 *gal*
Application event totals		59.71	12.62	78.30	405.57	

7 - 06/11/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.95	18.39	83.82	432.60	6,912,000.00 *gal*
River Water	Surface water	0.58	0.00	0.00	60.84	19,440,000.00 *gal*
Application event totals		69.53	18.39	83.82	493.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.71	13.79	62.86	324.45	5,184,000.00 *gal*
River Water	Surface water	0.58	0.00	0.00	60.84	19,440,000.00 *gal*
Application event totals		52.29	13.79	62.86	385.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.71	13.79	62.86	324.45	5,184,000.00 *gal*
River Water	Surface water	0.58	0.00	0.00	60.84	19,440,000.00 *gal*
Application event totals		52.29	13.79	62.86	385.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.96	12.26	55.88	288.40	4,608,000.00 *gal*
River Water	Surface water	0.58	0.00	0.00	60.84	19,440,000.00 *gal*
Application event totals		46.55	12.26	55.88	349.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.96	12.26	55.88	288.40	4,608,000.00 *gal*
River Water	Surface water	0.58	0.00	0.00	60.84	19,440,000.00 *gal*
Application event totals		46.55	12.26	55.88	349.24	

7 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.58	0.00	0.00	60.84	19,440,000.00 *gal*
Application event totals		0.58	0.00	0.00	60.84	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.58	0.00	0.00	60.84	19,440,000.00 *gal*
Application event totals		0.58	0.00	0.00	60.84	

8 - 11/22/2022: Winter Forage

Field name: 8

Crop: Winter Forage

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.83	9.90	61.41	318.09	720,000.00 *gal*
Application event totals		46.83	9.90	61.41	318.09	

01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.46	7.92	49.13	254.47	576,000.00 *gal*
Application event totals		37.46	7.92	49.13	254.47	

03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.46	7.92	49.13	254.47	576,000.00 *gal*
Application event totals		37.46	7.92	49.13	254.47	

8 - 06/16/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.48	15.86	72.31	373.23	3,168,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.89	11,016,000.00 *gal*
Application event totals		60.11	15.86	72.31	438.12	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.08	14.42	65.74	339.30	2,880,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.89	11,016,000.00 *gal*
Application event totals		54.70	14.42	65.74	404.19	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.08	14.42	65.74	339.30	2,880,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.89	11,016,000.00 *gal*
Application event totals		54.70	14.42	65.74	404.19	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.67	12.98	59.17	305.37	2,592,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.89	11,016,000.00 *gal*
Application event totals		49.29	12.98	59.17	370.26	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.67	12.98	59.17	305.37	2,592,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.89	11,016,000.00 *gal*
Application event totals		49.29	12.98	59.17	370.26	

8 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.67	12.98	59.17	305.37	2,592,000.00 *gal*
River Water	Surface water	0.62	0.00	0.00	64.89	11,016,000.00 *gal*
Application event totals		49.29	12.98	59.17	370.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.62	0.00	0.00	64.89	11,016,000.00 *gal*
Application event totals		0.62	0.00	0.00	64.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.62	0.00	0.00	64.89	11,016,000.00 *gal*
Application event totals		0.62	0.00	0.00	64.89	

9 - 11/22/2022: Winter Forage

Field name: 9

Crop: Winter Forage

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.61	11.96	74.24	384.54	576,000.00 *gal*
Application event totals		56.61	11.96	74.24	384.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.61	11.96	74.24	384.54	576,000.00 *gal*
Application event totals		56.61	11.96	74.24	384.54	

9 - 11/22/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	42.46	8.97	55.68	288.40	432,000.00 *gal*
Application event totals		42.46	8.97	55.68	288.40	

9 - 06/09/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.20	15.25	69.54	358.90	2,016,000.00 *gal*
River Water	Surface water	0.50	0.00	0.00	51.91	5,832,000.00 *gal*
Application event totals		57.70	15.25	69.54	410.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.03	13.07	59.60	307.63	1,728,000.00 *gal*
River Water	Surface water	0.50	0.00	0.00	51.91	5,832,000.00 *gal*
Application event totals		49.53	13.07	59.60	359.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.03	13.07	59.60	307.63	1,728,000.00 *gal*
River Water	Surface water	0.50	0.00	0.00	51.91	5,832,000.00 *gal*
Application event totals		49.53	13.07	59.60	359.54	

9 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.77	9.81	44.70	230.72	1,296,000.00 *gal*
River Water	Surface water	0.50	0.00	0.00	51.91	5,832,000.00 *gal*
Application event totals		37.27	9.81	44.70	282.64	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.77	9.81	44.70	230.72	1,296,000.00 *gal*
River Water	Surface water	0.50	0.00	0.00	51.91	5,832,000.00 *gal*
Application event totals		37.27	9.81	44.70	282.64	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.77	9.81	44.70	230.72	1,296,000.00 *gal*
River Water	Surface water	0.50	0.00	0.00	51.91	5,832,000.00 *gal*
Application event totals		37.27	9.81	44.70	282.64	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.50	0.00	0.00	51.91	5,832,000.00 *gal*
Application event totals		0.50	0.00	0.00	51.91	

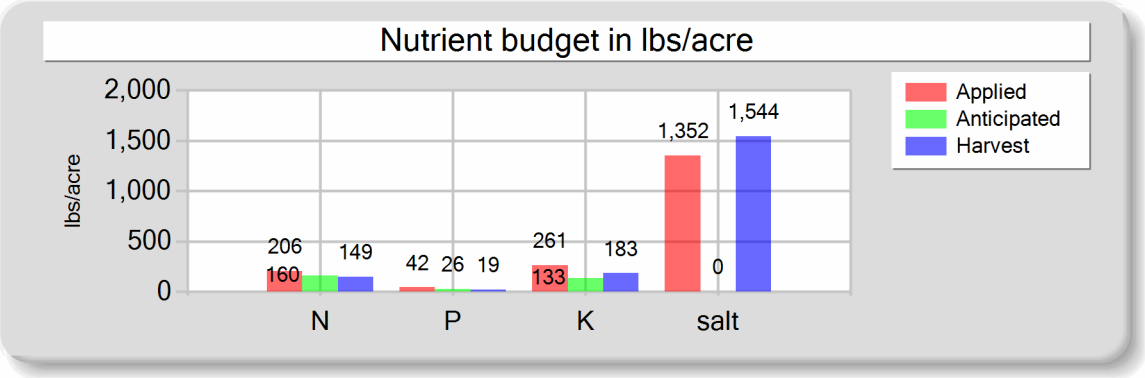
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River Water	Surface water	0.50	0.00	0.00	51.91	5,832,000.00 *gal*
Application event totals		0.50	0.00	0.00	51.91	

B. NUTRIENT BUDGET

1 - 11/28/2022: Winter Forage

Field name: 1 Crop: Winter Forage Plant date: 11/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.03	42.06	260.99	1,351.89
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.03	42.06	260.99	1,351.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	149.02	18.87	183.33	1,543.85
Nutrient balance	57.00	23.19	77.66	-191.96
Applied to removed ratio	1.38	2.23	1.42	0.88

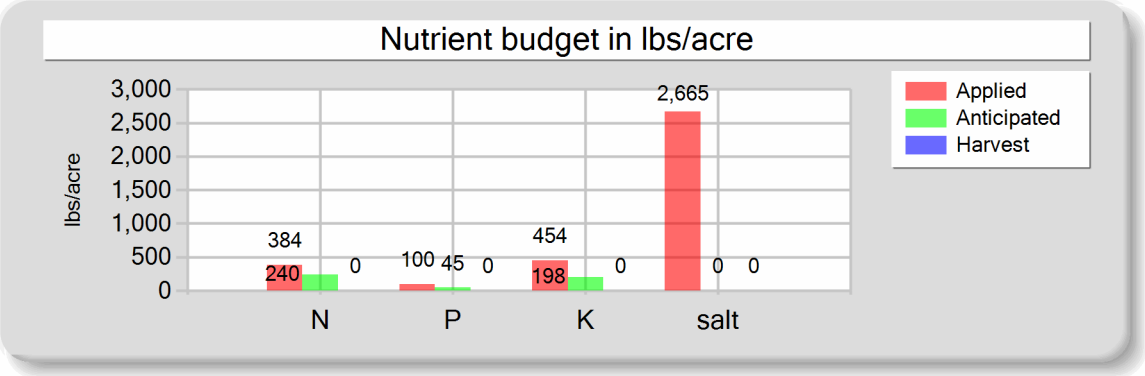
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
720,000.00	*gallons*
26.52	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/15/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	373.46	99.59	454.01	2,343.28
Fresh water	3.08	0.00	0.00	321.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	383.54	99.59	454.01	2,665.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	383.54	99.59	454.01	2,665.03
Applied to removed ratio	0.00	0.00	0.00	0.00

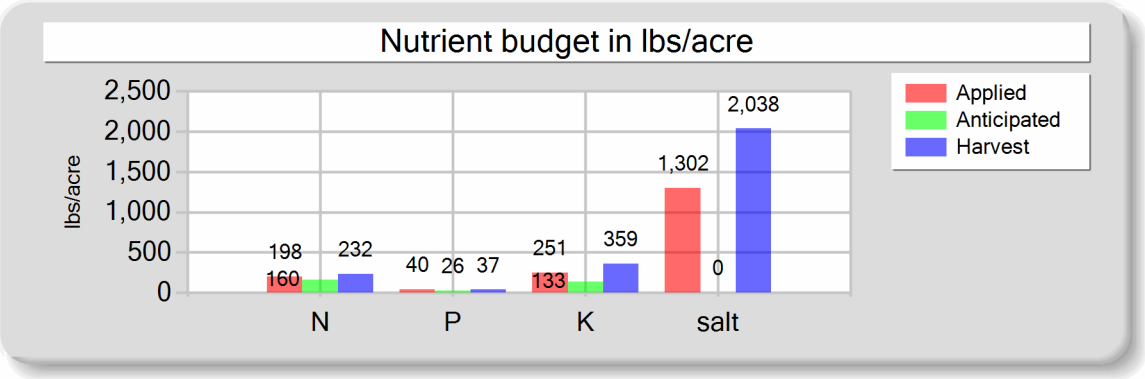
Fresh water applied	
12,852,000.00	*gallons*
473.30	*acre-inches*
29.58	*inches/acre*

Process wastewater applied	
4,680,000.00	*gallons*
172.35	*acre-inches*
10.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/28/2022: Winter Forage

Field name: 2 Crop: Winter Forage Plant date: 11/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.07	40.38	250.55	1,297.81
Fresh water	0.04	0.00	0.00	4.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.11	40.38	250.55	1,302.14
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	232.40	36.75	359.47	2,037.76
Nutrient balance	-34.29	3.63	-108.92	-735.62
Applied to removed ratio	0.85	1.10	0.70	0.64

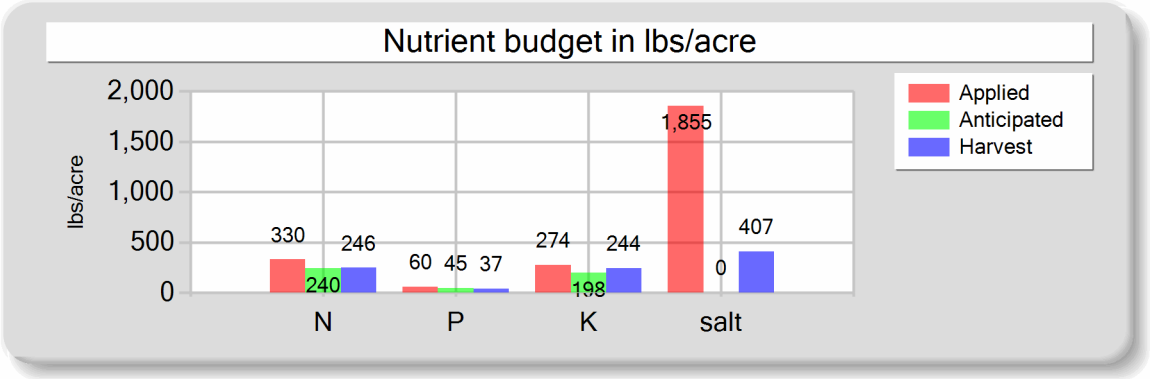
Fresh water applied	
540,000.00	*gallons*
19.89	*acre-inches*
0.40	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/15/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	318.59	60.47	274.03	1,440.09
Fresh water	3.98	0.00	0.00	415.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	329.57	60.47	274.03	1,855.39
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	245.94	37.36	244.06	407.40
Nutrient balance	83.63	23.11	29.97	1,448.00
Applied to removed ratio	1.34	1.62	1.12	4.55

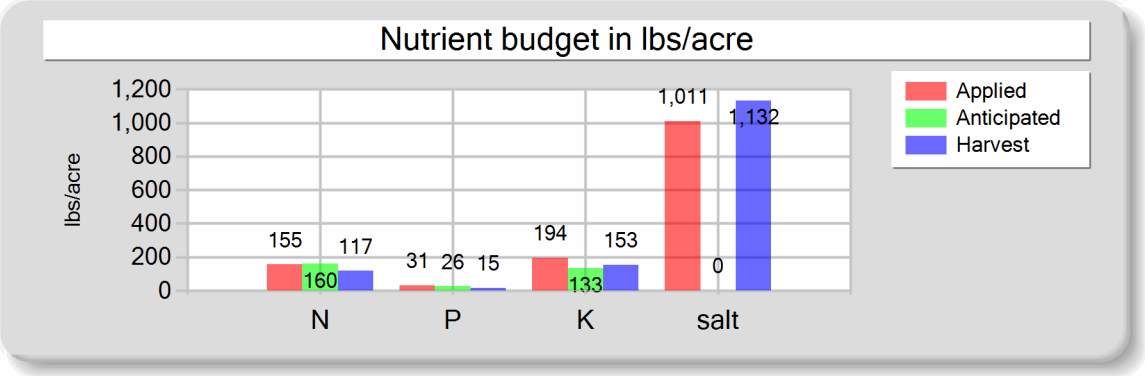
Fresh water applied	
51,840,000.00	*gallons*
1,909.09	*acre-inches*
38.18	*inches/acre*

Process wastewater applied	
8,208,000.00	*gallons*
302.27	*acre-inches*
6.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/29/2022: Winter Forage

Field name: 3 Crop: Winter Forage Plant date: 11/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	148.11	31.30	194.23	1,006.06
Fresh water	0.05	0.00	0.00	5.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	155.16	31.30	194.23	1,011.09
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	117.30	14.66	153.15	1,132.31
Nutrient balance	37.86	16.64	41.08	-121.22
Applied to removed ratio	1.32	2.13	1.27	0.89

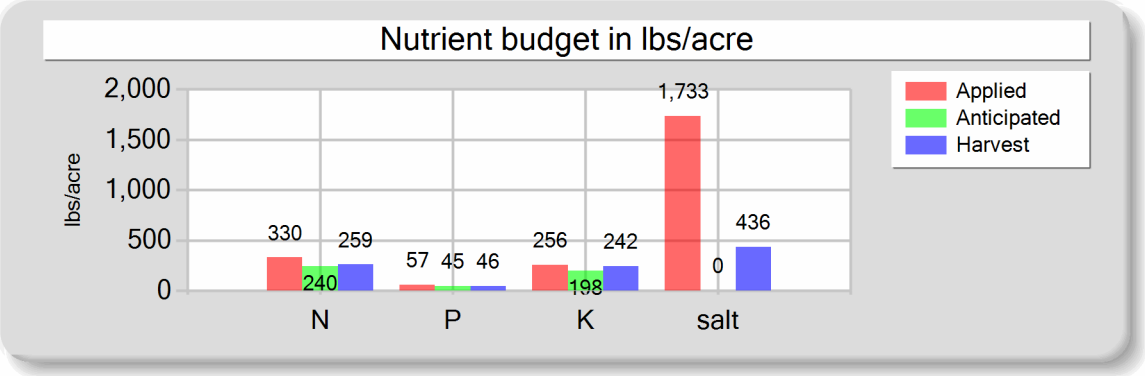
Fresh water applied	
540,000.00	*gallons*
19.89	*acre-inches*
0.46	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/16/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	319.14	56.63	256.27	1,352.59
Fresh water	3.64	0.00	0.00	380.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	329.79	56.63	256.27	1,732.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	258.62	45.95	241.52	436.03
Nutrient balance	71.16	10.68	14.74	1,296.85
Applied to removed ratio	1.28	1.23	1.06	3.97

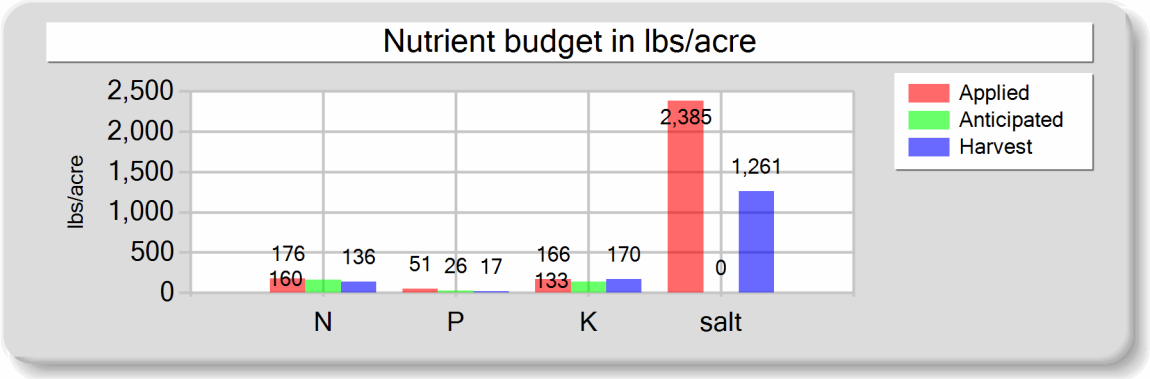
Fresh water applied	
40,824,000.00	*gallons*
1,503.41	*acre-inches*
34.96	*inches/acre*

Process wastewater applied	
6,480,000.00	*gallons*
238.64	*acre-inches*
5.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/22/2022: Winter Forage

Field name: 4 Crop: Winter Forage Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.83	42.81	111.30	2,132.37
Process wastewater	41.51	8.65	54.75	252.99
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.35	51.46	166.05	2,385.36
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	135.77	16.78	169.72	1,260.74
Nutrient balance	40.57	34.69	-3.67	1,124.62
Applied to removed ratio	1.30	3.07	0.98	1.89

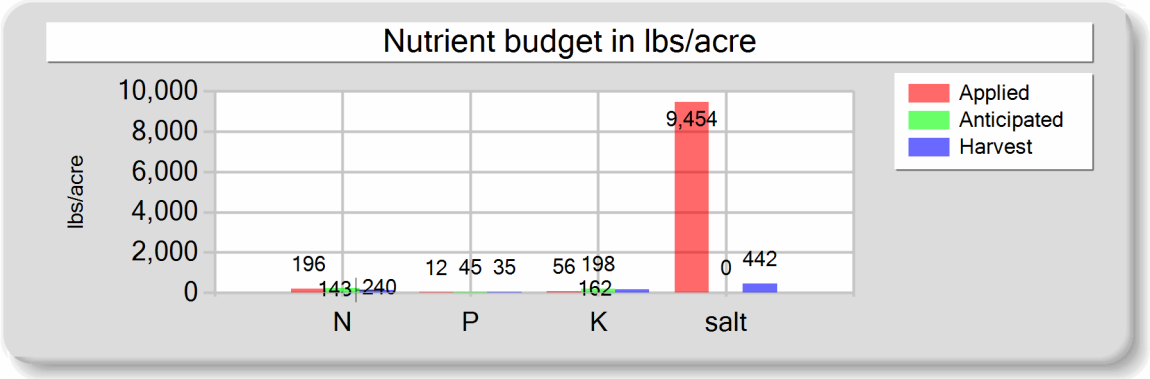
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
864,000.00 *gallons*
31.82 *acre-inches*
0.33 *inches/acre*

Total harvests for the crop
1 *harvests*

4 - 06/19/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	46.45	12.39	56.47	291.44
Fresh water	143.02	0.00	0.00	9,162.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.47	12.39	56.47	9,453.72
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	142.65	35.46	161.60	442.22
Nutrient balance	53.82	-23.07	-105.14	9,011.50
Applied to removed ratio	1.38	0.35	0.35	21.38

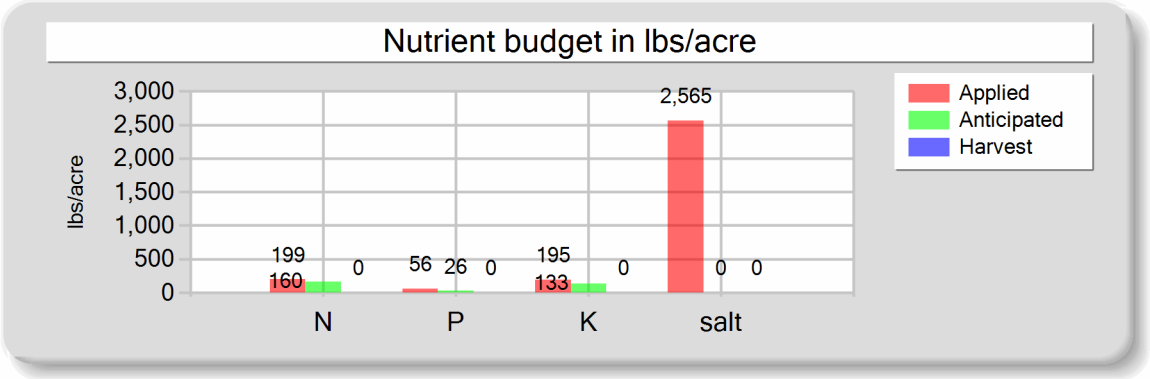
Fresh water applied	
50,880,000.00	*gallons*
1,873.74	*acre-inches*
19.72	*inches/acre*

Process wastewater applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

4B - 11/21/2022: Winter Forage

Field name: 4B | Crop: Winter Forage | Plant date: 11/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.83	42.81	111.30	2,132.37
Process wastewater	63.69	13.46	83.52	432.60
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.52	56.27	194.82	2,564.97
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	198.52	56.27	194.82	2,564.97
Applied to removed ratio	0.00	0.00	0.00	0.00

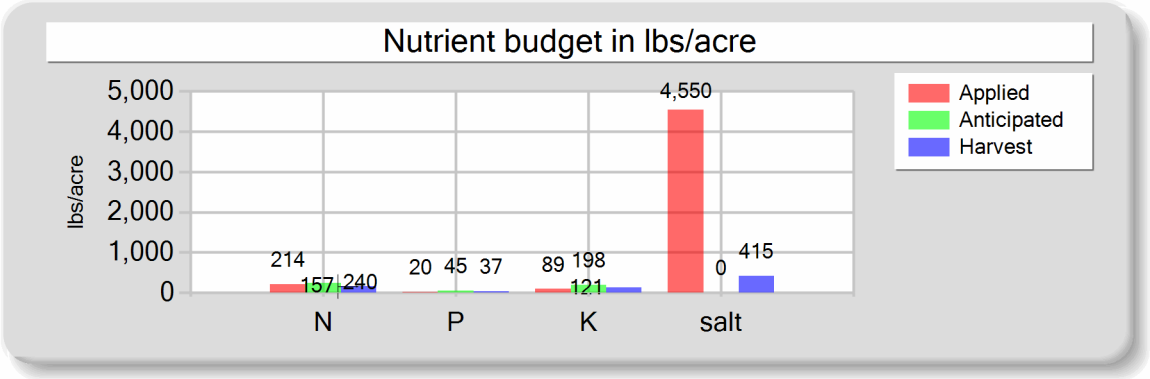
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
0.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

4B - 06/19/2023: Corn, silage

Field name: 4B Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	73.54	19.61	89.40	461.45
Fresh water	133.72	0.00	0.00	4,088.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.26	19.61	89.40	4,549.56
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	156.69	37.40	121.07	415.38
Nutrient balance	57.57	-17.79	-31.66	4,134.18
Applied to removed ratio	1.37	0.52	0.74	10.95

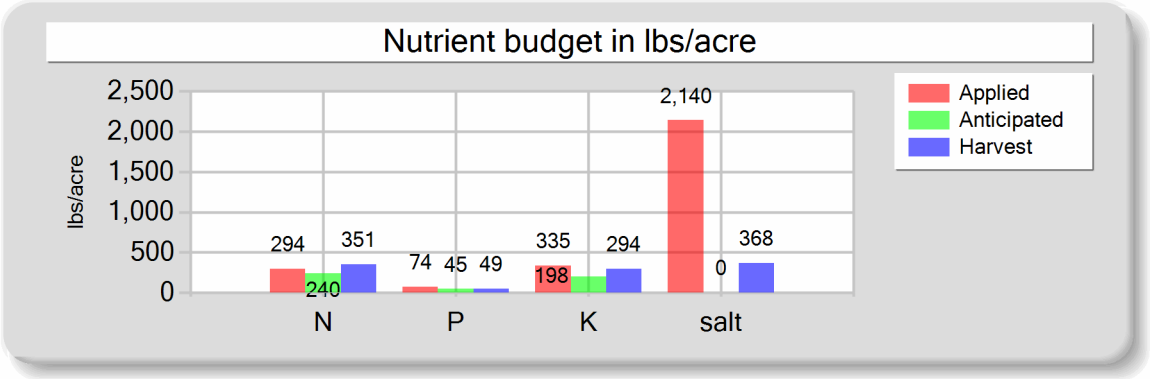
Fresh water applied	
40,824,000.00	*gallons*
1,503.41	*acre-inches*
37.59	*inches/acre*

Process wastewater applied	
2,304,000.00	*gallons*
84.85	*acre-inches*
2.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

5A - 06/13/2023: Corn, silage

Field name: 5A | Crop: Corn, silage | Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.79	73.54	335.27	1,730.42
Fresh water	3.93	0.00	0.00	409.84
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	293.71	73.54	335.27	2,140.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	350.64	48.75	294.39	367.52
Nutrient balance	-56.93	24.79	40.88	1,772.74
Applied to removed ratio	0.84	1.51	1.14	5.82

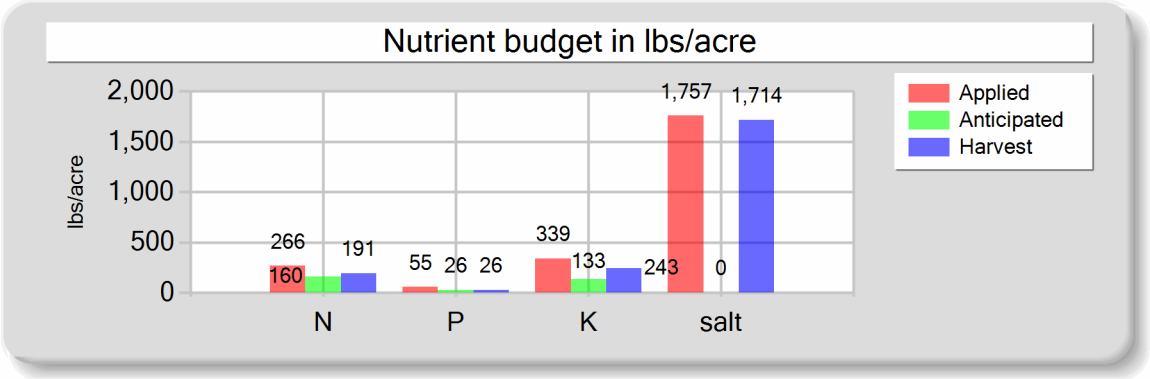
Fresh water applied	
38,880,000.00	*gallons*
1,431.82	*acre-inches*
37.68	*inches/acre*

Process wastewater applied	
8,208,000.00	*gallons*
302.27	*acre-inches*
7.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

5B - 11/23/2022: Winter Forage

Field name: 5B Crop: Winter Forage Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	258.74	54.68	339.29	1,757.46
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.74	54.68	339.29	1,757.46
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	190.96	26.07	242.53	1,714.44
Nutrient balance	74.78	28.61	96.76	43.02
Applied to removed ratio	1.39	2.10	1.40	1.03

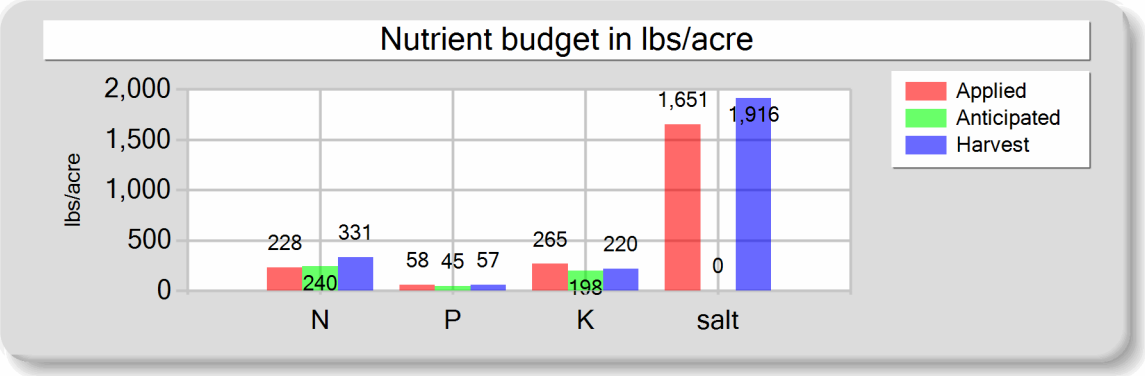
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
936,000.00	*gallons*
34.47	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

5B - 06/13/2023: Corn, silage

Field name: 5B | Crop: Corn, silage | Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.33	58.22	265.42	1,369.92
Fresh water	2.69	0.00	0.00	281.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.03	58.22	265.42	1,651.11
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	330.98	56.90	219.67	1,915.62
Nutrient balance	-102.96	1.32	45.75	-264.51
Applied to removed ratio	0.69	1.02	1.21	0.86

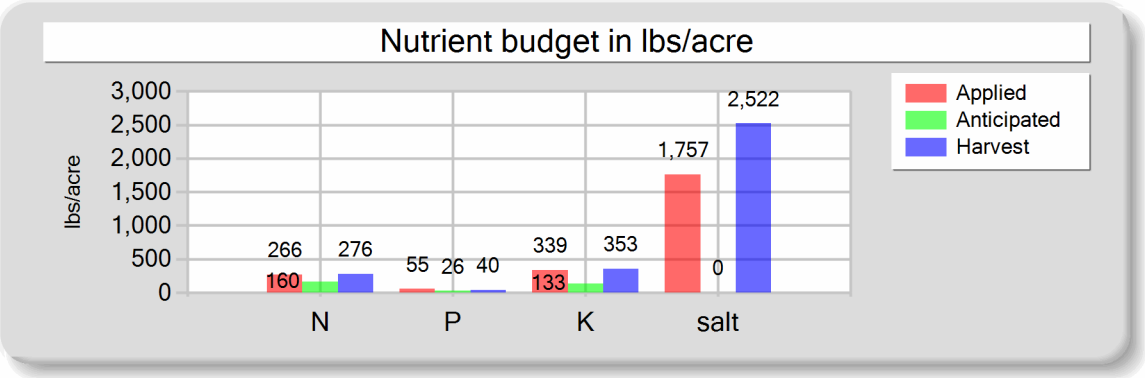
Fresh water applied	
11,232,000.00	*gallons*
413.64	*acre-inches*
25.85	*inches/acre*

Process wastewater applied	
2,736,000.00	*gallons*
100.76	*acre-inches*
6.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

5C - 11/17/2022: Winter Forage

Field name: 5C | Crop: Winter Forage | Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	258.74	54.68	339.29	1,757.46
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.74	54.68	339.29	1,757.46
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	275.91	39.65	352.66	2,521.60
Nutrient balance	-10.18	15.03	-13.37	-764.14
Applied to removed ratio	0.96	1.38	0.96	0.70

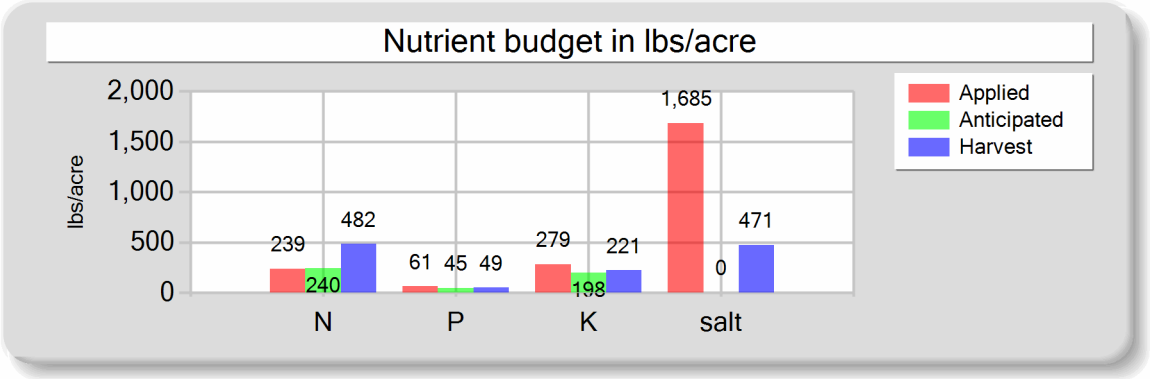
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
936,000.00	*gallons*
34.47	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

5C - 06/14/2023: Corn, silage

Field name: 5C | Crop: Corn, silage | Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.82	61.29	279.39	1,442.02
Fresh water	2.33	0.00	0.00	243.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.15	61.29	279.39	1,685.36
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	482.05	48.66	221.15	470.62
Nutrient balance	-242.89	12.62	58.24	1,214.73
Applied to removed ratio	0.50	1.26	1.26	3.58

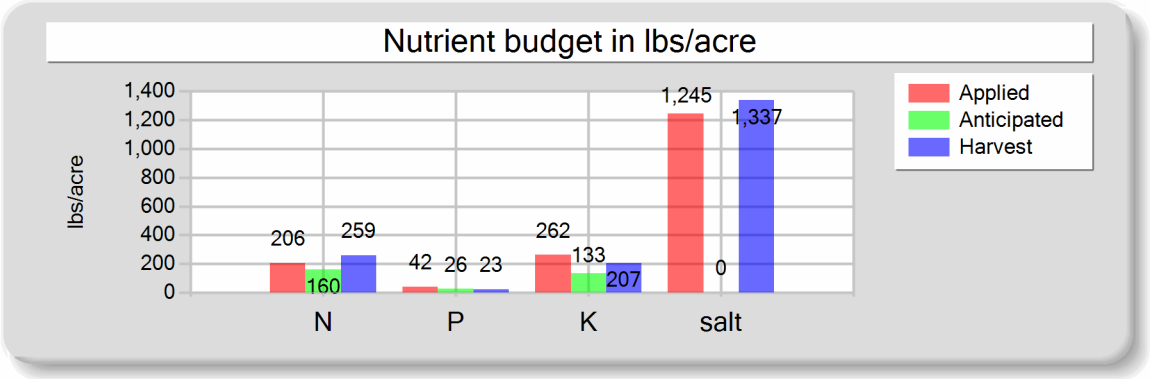
Fresh water applied	
9,720,000.00	*gallons*
357.95	*acre-inches*
22.37	*inches/acre*

Process wastewater applied	
2,880,000.00	*gallons*
106.06	*acre-inches*
6.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/30/2022: Winter Forage

Field name: 6 Crop: Winter Forage Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	198.77	41.57	261.76	1,245.17
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.77	41.57	261.76	1,245.17
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	259.21	22.78	206.90	1,336.73
Nutrient balance	-53.44	18.78	54.86	-91.56
Applied to removed ratio	0.79	1.82	1.27	0.93

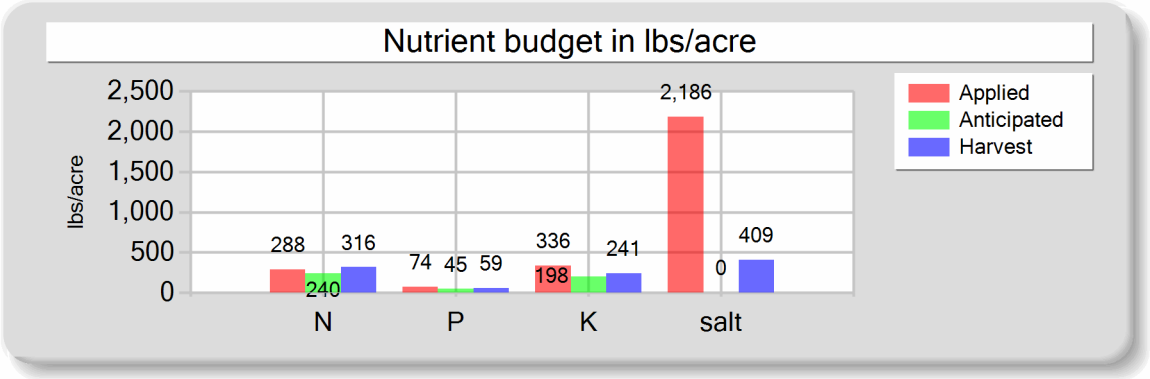
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
4,608,000.00 *gallons*
169.70 *acre-inches*
1.62 *inches/acre*

Total harvests for the crop
1 *harvests*

6 - 06/13/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	276.66	73.78	336.33	1,735.91
Fresh water	4.31	0.00	0.00	449.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.97	73.78	336.33	2,185.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	316.45	59.30	240.70	408.64
Nutrient balance	-28.48	14.47	95.63	1,777.18
Applied to removed ratio	0.91	1.24	1.40	5.35

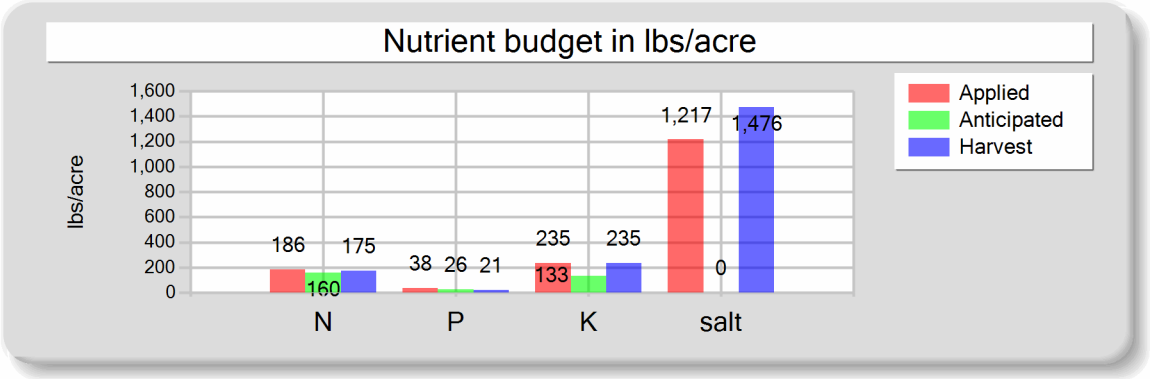
Fresh water applied	
117,936,000.00	*gallons*
4,343.18	*acre-inches*
41.36	*inches/acre*

Process wastewater applied	
22,752,000.00	*gallons*
837.88	*acre-inches*
7.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/18/2022: Winter Forage

Field name: 7 Crop: Winter Forage Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	179.13	37.85	234.89	1,216.70
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	186.13	37.85	234.89	1,216.70
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	175.47	20.99	234.50	1,475.88
Nutrient balance	10.66	16.86	0.39	-259.18
Applied to removed ratio	1.06	1.80	1.00	0.82

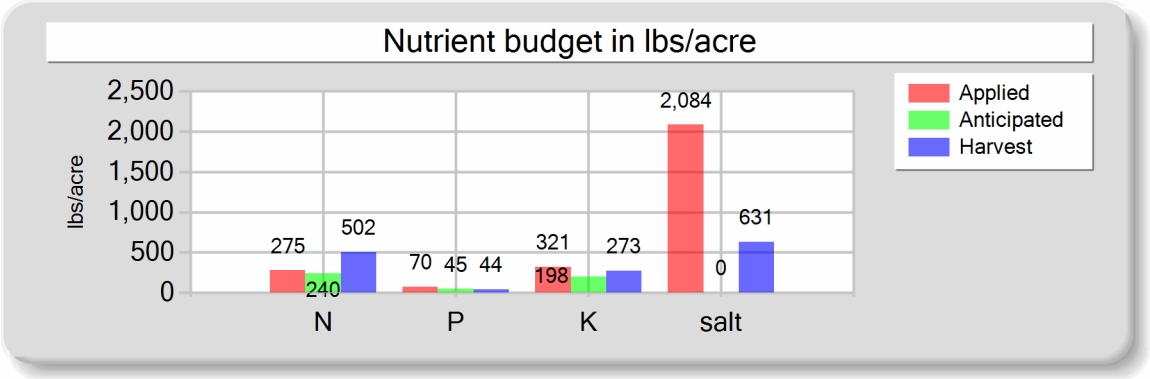
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
5,184,000.00 *gallons*
190.91 *acre-inches*
1.49 *inches/acre*

Total harvests for the crop
1 *harvests*

7 - 06/11/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	264.29	70.48	321.30	1,658.32
Fresh water	4.08	0.00	0.00	425.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.38	70.48	321.30	2,084.16
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	502.43	43.62	272.63	630.95
Nutrient balance	-227.05	26.86	48.66	1,453.21
Applied to removed ratio	0.55	1.62	1.18	3.30

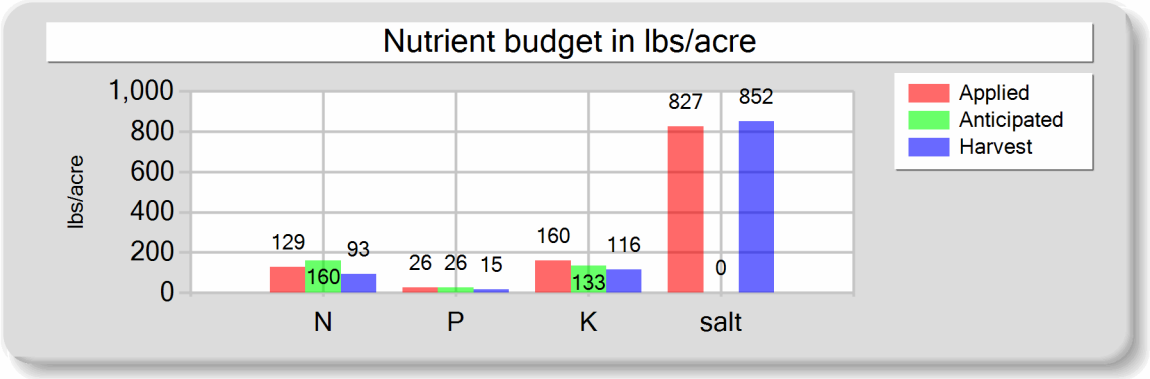
Fresh water applied	
136,080,000.00	*gallons*
5,011.36	*acre-inches*
39.15	*inches/acre*

Process wastewater applied	
26,496,000.00	*gallons*
975.76	*acre-inches*
7.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/22/2022: Winter Forage

Field name: 8 Crop: Winter Forage Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	121.76	25.73	159.66	827.04
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	128.76	25.73	159.66	827.04
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	92.91	15.47	115.53	851.76
Nutrient balance	35.85	10.26	44.14	-24.72
Applied to removed ratio	1.39	1.66	1.38	0.97

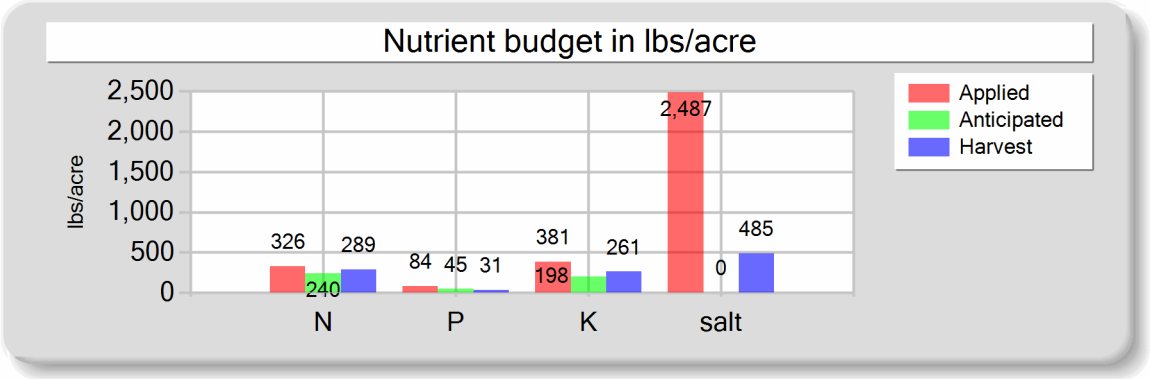
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,872,000.00	*gallons*
68.94	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 06/16/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	313.64	83.64	381.29	1,967.93
Fresh water	4.97	0.00	0.00	519.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.61	83.64	381.29	2,487.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	288.92	30.71	260.64	484.91
Nutrient balance	36.69	52.93	120.65	2,002.15
Applied to removed ratio	1.13	2.72	1.46	5.13

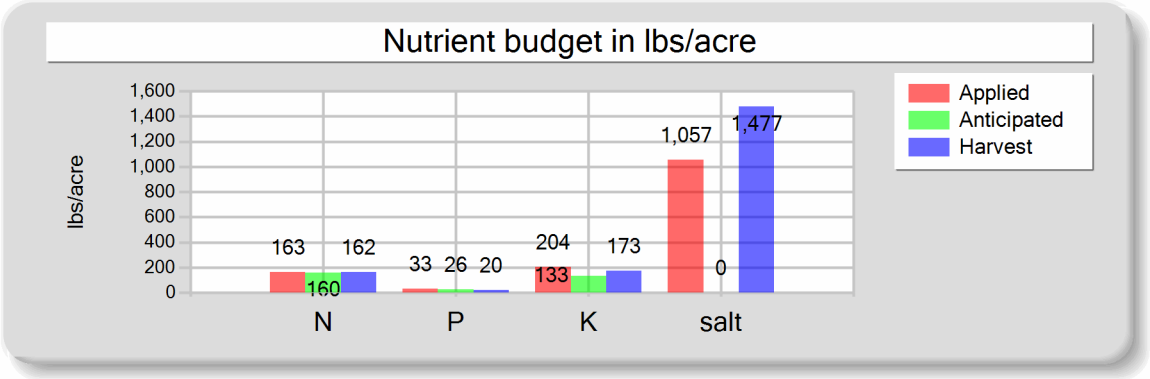
Fresh water applied	
88,128,000.00	*gallons*
3,245.45	*acre-inches*
47.73	*inches/acre*

Process wastewater applied	
16,704,000.00	*gallons*
615.15	*acre-inches*
9.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 11/22/2022: Winter Forage

Field name: 9 Crop: Winter Forage Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	155.68	32.90	204.15	1,057.48
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	162.68	32.90	204.15	1,057.48
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	162.23	19.52	173.44	1,477.22
Nutrient balance	0.45	13.38	30.71	-419.74
Applied to removed ratio	1.00	1.69	1.18	0.72

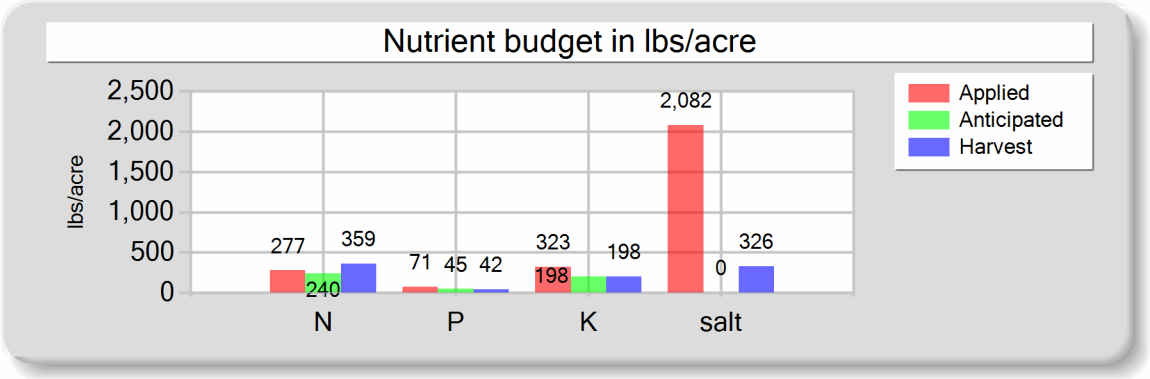
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,584,000.00	*gallons*
58.33	*acre-inches*
1.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 06/09/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	265.57	70.82	322.85	1,666.33
Fresh water	3.98	0.00	0.00	415.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	276.55	70.82	322.85	2,081.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	358.86	42.11	197.74	325.91
Nutrient balance	-82.31	28.71	125.11	1,755.73
Applied to removed ratio	0.77	1.68	1.63	6.39

Fresh water applied	
46,656,000.00	*gallons*
1,718.18	*acre-inches*
38.18	*inches/acre*

Process wastewater applied	
9,360,000.00	*gallons*
344.70	*acre-inches*
7.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Compost/ Manure 2092312-01

Sample and source description: Compost/ Manure 2092312-01

Sample date: 09/22/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	7,770.00	20,200.00	8,700.00	6,400.00	2,100.00	1,990.00	3,400.00		38.70
DL	0.50	0.50	0.50	5.00	5.00	5.00	1.00	0.50		0.10

HC/Manure 2092312-02

Sample and source description: HC/Manure 2092312-02

Sample date: 09/22/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting:

Moisture: 47.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value										
DL										

HCX/Manure 2307238-02

Sample and source description: HCX/Manure 2307238-02

Sample date: 07/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting:

Moisture: 36.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value										
DL										

MMC/Manure 2307238-01

Sample and source description: MMC/Manure 2307238-01

Sample date: 07/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting:

Moisture: 22.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value										
DL										

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302051-01

Sample and source description: Lagoon Water 2302051-01

Sample date: 01/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	530.00	530.00			112.00	695.00								4,590.00	3,600
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303400-01

Sample and source description: Lagoon Water 2303400-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	581.00	581.00			118.00	774.00								5,030.00	2,800
DL	0.60	10.00			0.10	10.00								1.00	1

Lagoon Water 2307237-01

Sample and source description: Lagoon Water 2307237-01

Sample date: 07/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	153.00	138.00			40.80	186.00								2,060.00	960
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310020-01

Sample and source description: Lagoon Water 2310020-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,060.00	212.00			78.90	346.00								3,230.00	2,000
DL	3.00	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well #8

2312510-08

Sample description: 2312510-08

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.95										233.00	140
DL	1.40										1.00	1

Ag Well 4A East South

Ag Well 4A East South

2312510-06

Sample description: 2312510-06

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	35.80										1,490.00	1,100
DL	1.40										1.00	1

Ag Well 4A West

2312510-05

Sample description: 2312510-05

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	28.20										1,480.00	3,000
DL	1.40										1.00	1

Ag Well 4B

2312510-07

Sample description: 2312510-07

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	15.70										678.00	480
DL	1.40										1.00	1

Canal Water

Canal Water

2312510-09

Sample description: 2312510-09

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											42.00	26
DL											1.00	1

Domestic Well #1

2312510-01

Sample description: 2312510-01

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.23								144.00	
DL			0.40								1.00	

Domestic Well #2

2312510-02

Sample description: 2312510-02

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.24								149.00	
DL			0.40								1.00	

Domestic Well #3 by Corral

Domestic Well #3 by Corral

2312510-03

Sample description: 2312510-03

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.25								149.00	
DL			0.40								1.00	

Domestic well #4 by Calf

2312510-04

Sample description: 2312510-04

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			6.09								278.00	
DL			0.40								1.00	

River Water

River Water 2310021-01

Sample description: River Water 2310021-01

Sample date: 09/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.46										33.20	48
DL	0.40										1.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/28/2022: Winter Forage

2305489-01

Sample and source description: 2305489-01

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	1,760.00	17,100.00		14.40
DL	0.50	0.50	0.50		0.10

2 - 11/28/2022: Winter Forage

2305241-01

Sample and source description: 2305241-01

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	3,210.00	31,400.00		17.80
DL	1.00	0.50	0.50		0.10

2 - 06/15/2023: Corn, silage

2 - 06/15/2023: Corn, silage

2310279-01

Sample and source description: 2310279-01

Sample date: 10/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	1,990.00	13,000.00		2.17
DL	0.50	0.50	0.50		0.10

3 - 11/29/2022: Winter Forage

2305489-02

Sample and source description: 2305489-02

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	1,800.00	18,800.00		13.90
DL	0.50	0.50	0.50		0.10

3 - 06/16/2023: Corn, silage

2310279-02

Sample and source description: 2310279-02

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,150.00	11,300.00		2.04
DL	0.50	0.50	0.50		0.10

4 - 11/22/2022: Winter Forage

2305489-03

Sample and source description: 2305489-03

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	1,730.00	17,500.00		13.00
DL	0.50	0.50	0.50		0.10

4 - 06/19/2023: Corn, silage

2310279-03

Sample and source description: 2310279-03

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,000.00	1,740.00	7,930.00		2.17
DL	0.50	0.50	0.50		0.10

4B - 06/19/2023: Corn, silage

2310279-04

Sample and source description: 2310279-04

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	2,530.00	8,190.00		2.81
DL	0.50	0.50	0.50		0.10

5A - 06/13/2023: Corn, silage

2310279-05

Sample and source description: 2310279-05

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,700.00	2,600.00	15,700.00		1.96
DL	1.00	0.50	0.50		0.10

5B - 11/23/2022: Winter Forage

2305489-04

Sample and source description: 2305489-04

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	1,870.00	17,400.00		12.30
DL	1.00	0.50	0.50		0.10

5B - 06/13/2023: Corn, silage

2310279-06

Sample and source description: 2310279-06

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,510.00	9,690.00		8.45
DL	0.50	0.50	0.50		0.10

5C - 11/17/2022: Winter Forage

2305489-05

Sample and source description: 2305489-05

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	2,170.00	19,300.00		13.80
DL	1.00	0.50	0.50		0.10

5C - 06/14/2023: Corn, silage

2310279-07

Sample and source description: 2310279-07

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,100.00	2,130.00	9,680.00		2.06
DL	0.50	0.50	0.50		0.10

6 - 11/30/2022: Winter Forage

2305489-06

Sample and source description: 2305489-06

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	1,960.00	17,800.00		11.50
DL	2.50	0.50	0.50		0.10

6 - 06/13/2023: Corn, silage

2310279-08

Sample and source description: 2310279-08

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,380.00	9,660.00		1.64
DL	0.50	0.50	0.50		0.10

7 - 11/18/2022: Winter Forage

2305489-07

Sample and source description: 2305489-07

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	1,280.00	14,300.00		9.00
DL	0.50	0.50	0.50		0.10

7 - 06/11/2023: Corn, silage

2310279-09

Sample and source description: 2310279-09

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,800.00	2,240.00	14,000.00		3.24
DL	0.50	0.50	0.50		0.10

8 - 11/22/2022: Winter Forage

2305489-09

Sample and source description: 2305489-09

Sample date: 05/30/2022 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,490.00	1,580.00	11,800.00		8.70
DL	0.50	0.50	0.50		0.10

2305489-08

Sample and source description: 2305489-08

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	2,360.00	22,100.00		14.40
DL	1.00	0.50	0.50		0.10

8 - 06/16/2023: Corn, silage

2310279-10

Sample and source description: 2310279-10

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	1,520.00	12,900.00		2.40
DL	0.50	0.50	0.50		0.10

9 - 11/22/2022: Winter Forage

9 - 11/22/2022: Winter Forage

2305241-03

Sample and source description: 2305241-03

Sample date: 05/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,600.00	2,960.00	26,300.00		22.40
DL	2.50	0.50	0.50		0.10

9 - 06/09/2023: Corn, silage

2310279-11

Sample and source description: 2310279-11

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	2,300.00	10,800.00		1.78
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

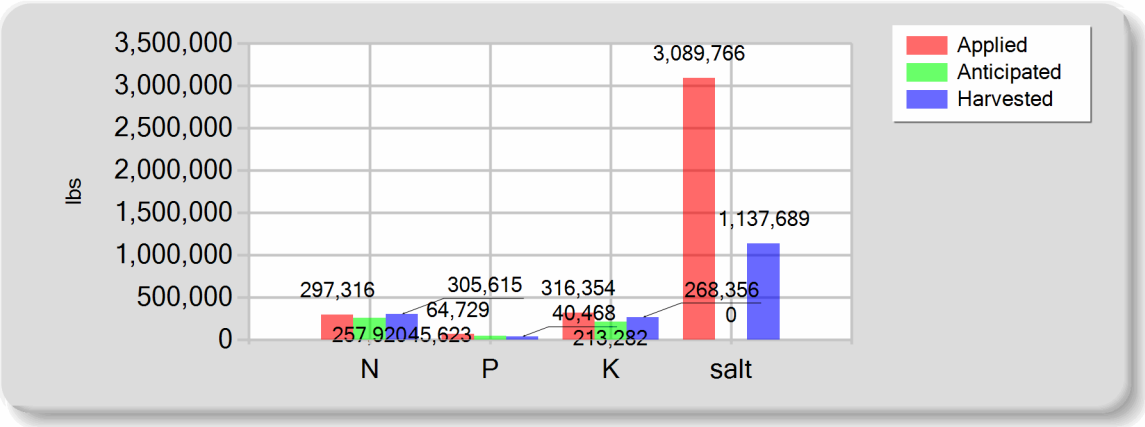
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

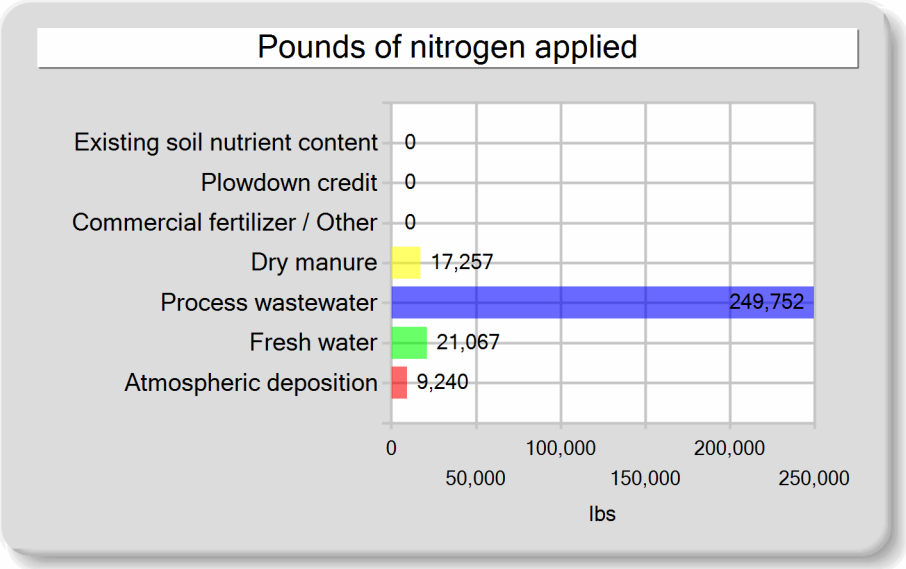
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

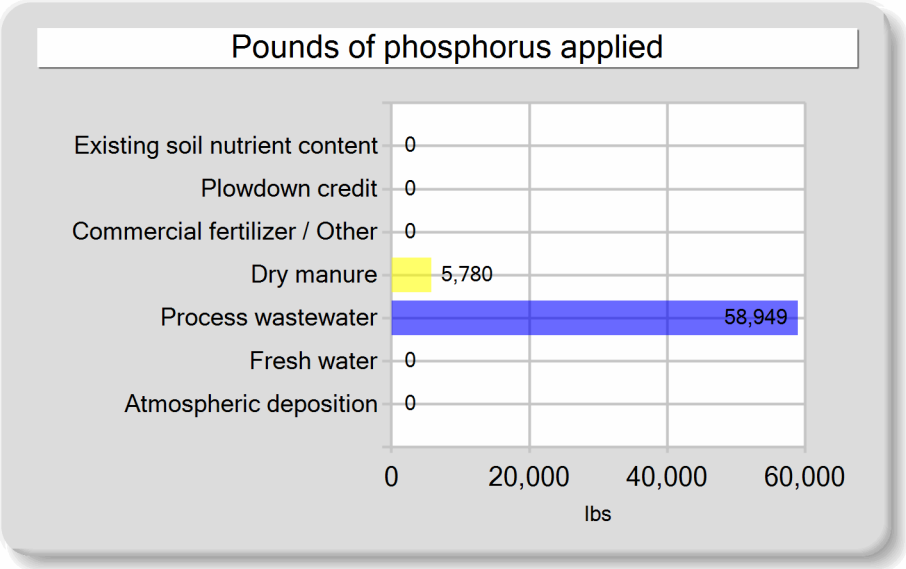
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	17,257.32	5,779.71	15,025.77	287,869.95
Process wastewater	249,751.76	58,949.37	301,328.47	1,545,552.75
Fresh water	21,066.97	0.00	0.00	1,256,343.62
Atmospheric deposition	9,240.00	0.00	0.00	0.00
Total nutrients applied	297,316.05	64,729.09	316,354.24	3,089,766.32
Anticipated crop nutrient removal	257,920.00	45,623.20	213,281.60	0.00
Actual crop nutrient removal	305,614.92	40,468.28	268,355.58	1,137,688.94
Nutrient balance	-8,298.87	24,260.81	47,998.66	1,952,077.38
Applied to removed ratio	0.97	1.60	1.18	2.72

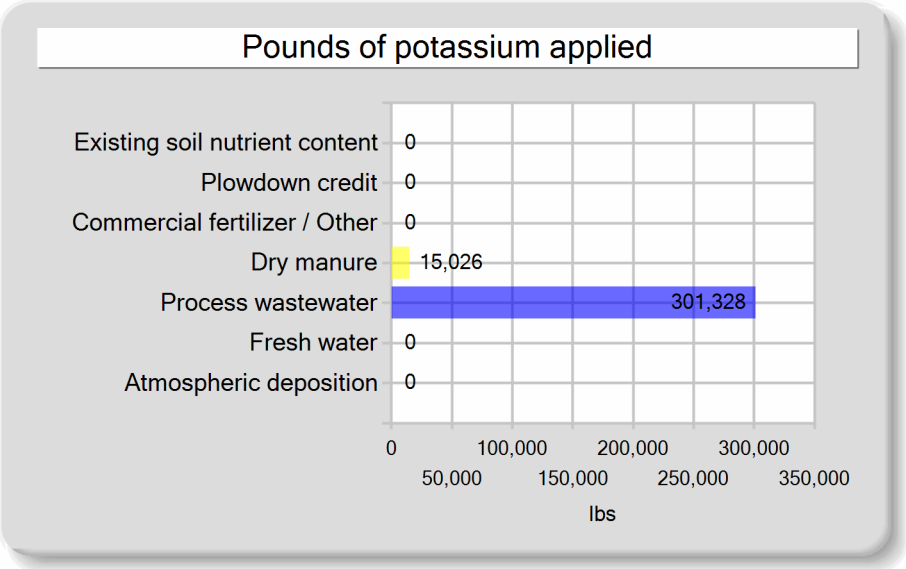
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

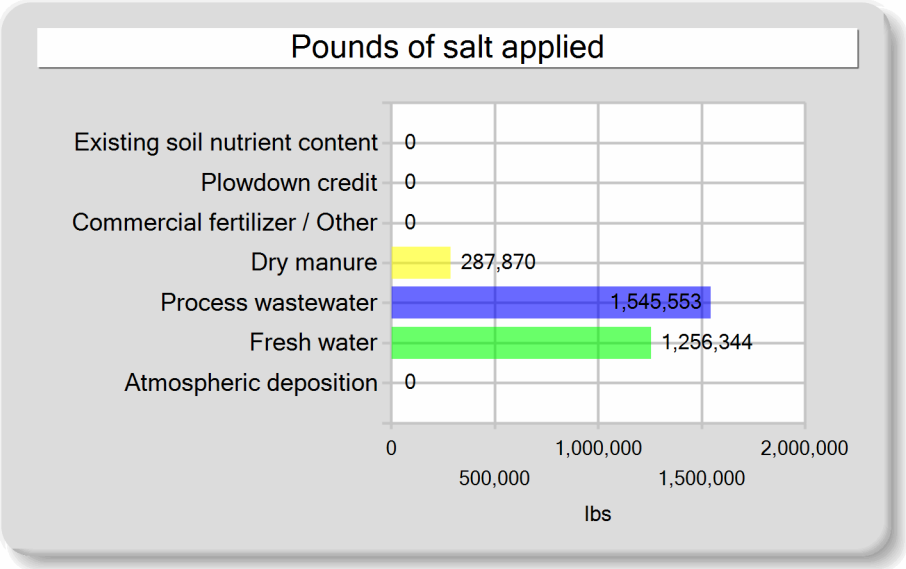


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

*Field 4B Winter Forage was not sampled in 2023, we will be sure to sample in 2024.

Field 1 Corn Silage was not sampled in 2023, we will be sure to sample in 2024.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X	X
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jeanette Veldhuis	Paul Van Waremerdam
PRINT OR TYPE NAME	PRINT OR TYPE NAME
✓ Jeanette Veldhuis	✓ 6/27/24
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	See Attachments
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	